

Almat Abdimalik

Automation and Robotics Engineer

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Skills and Interests

Programming Languages: C++, Python, MATLAB, IEC 61131-3

Robotics: ROS1 Noetic, ROS2 Humble, Rviz2, MoveIt2, Computer Vision, Machine Learning, Isaac Sim

Automation & Control: Siemens TIA Portal, Schneider Control Expert, Autodesk Fusion 360, Unisim

Electronics and Hardware Skills: Raspberry Pi 4/5, ESP32, Nvidia Jetson, Intel RealSense D435i, Modicon M340/M241 PLCs, Simatic S7-1200/S7-300 PLCs, KUKA KRC4 compact

Experience

Engineer, Siemens Laboratory Sep 2025 – Present

- Managed setup, configuration, and maintenance of SIMATIC S7-1200/S7-1500 controllers, ET 200SP I/O, industrial networks, and PCS (PC) systems using Siemens software tools

Siemens Laboratory Intern Jul 2025 – Aug 2025

- Developed PLC projects in TIA Portal with SIMIT-based simulation and real-hardware testing on SIMATIC S7-1200/S7-1500
- Supported control tasks and system validation using MATLAB–PLC communication

Assistant, KBTU Robotics Laboratory Sep 2024 – May 2025

- Programmed Raspberry Pi and microcontrollers for robotics laboratory tasks
- Produced technical schematics in AutoCAD and designed 3D models in Fusion 360 for 3D printing

Projects

Development of multirobot painting control system with realtime monitoring

- Automated painting process, consists of painting robots, conveyor line, drying process, pick and place robot
- PLC-based architecture controlling conveyor belt and MATLAB controlling manipulators and real-time data figures
- Tools Used: KUKA, RoArm M2s, BLDC motor, MATLAB, Siemens SIMATIC S7-1500, WINCC (HMI)

Machine Learning and MATLAB in Industry

- Predict product quality in an industrial mining process with ML in MATLAB and connect to PLC by OPC UA communication
- Tools Used: MATLAB, Python, Machine Learning, Siemens SIMATIC S7-1200, HMI

Industrial Automation Internship Project at Hyundai Trans Kazakhstan

- PLC-based conveyor control, HMI development, and practical implementation of electrical and automation systems
- Tools Used: Siemens SIMATIC S7-300, Conveyor belt, LS Electric HMI, Proximity Sensors

PID-Based Self-Balancing Control of a BLDC Motor

- A feedback control project implementing a PID controller for a self-balancing BLDC motor system, with real-time regulation and stability-oriented tuning.
- Tools Used: C++, MATLAB/Simulink, BLDC Motor, Microcontroller, PID Control

Inverted Pendulum on a Moving Platform | Control Systems

- A self-balancing inverted pendulum system on a mobile platform, developed using an ESP32 microcontroller with PID-based feedback control and IMU-based angle estimation.
- Tools Used: ESP32, C++, PID Control, IMU6050, Complementary Filter, Fusion 360, MATLAB/Simulink

Education

Kazakh-British Technical University, BS in Automation and Control

Sept 2022 – May 2026

- **American accreditation ABET**
- **Coursework:** Foundations of Electrical Engineering, Theory of Linear and Non-linear Control Systems, Autonomous Mobile Robots, Industrial Robot Operations, Introduction to Machine Learning, Metrology and Measurement

Activities and Achievements

ASTP discipline: Won 2nd place for the best course project among 40+ teams from the entire 4th-year Automation and Control cohort at university

Mentor in Robo-Football Hackathon: Mentor for students and served as a judge in the Robo-Football Hackathon during ITFEST 2024 and ITFEST 2025

Open Day Robotics KBTU: Guided and advised prospective students on the Robotics and Automation program